Bd Genetics Analysis Plan

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Contents

1 Helpful Papers

Here are some papers that I found very helpful in the processes of coming up with this plan. I will attempt to cite which papers did what where but that may not be completely accurate during the main body of the text. Feel safe in knowing that these specific papers are the primary sources for all the information that I have compiled though!

- A beginner's guide to assembling a draft genome and analyzing structural variants with long-read sequencing technologies. [6]
- Complex history of amphibian killing fungus reveals new insights into pathogen genotype and virulence. [17]
- Chromosomal copy number variation in the amphibian killing fungus Batrachochytrium dendrobatidis. [13]
- Hybrid Assembly of the Chytrid Genome - Insights into the Evolution and Outbreak of a Major Amphibian Pathogen
- Diploidy within a Haploid Genus of Entomopathogenic Fungi. [14]
- Cryptic diversity of a widespread global pathogen reveals expanded threats to amphibian conservation. [4]
- Genomic Correlates of Virulence Attenuation in the Deadly Amphibian Chytrid Fungus, Batrachochytrium dendrobatidis [16]
- Chapter 8 - Decoding the blueprint: an overview of genome sequencing, assembly, and annotation [5]
- Genomic Perspectives on the Pathogenic Chytrid Fungus and Persisting Amphibian Hosts [3]

2 Introduction and preamble

There are a couple different approaches to using sequence data for my applications. For this work I have a few related but ultimately independent goals:

1. Create phylogenies with older NGS sequence data
    - Mitochondrial
    - Genomic DNA
2. Create phylogenies with new PacBio long read data
    - Mitochondrial
    - Genomic DNA
3. Create CCNV Estimates of existing NGS sequence data
4. Create CCNV Estimates of new PacBio long read data

3 Available Sequence Data

1. **Published NGS Sequence Data:** There are approximately 40 published Bd NGS sequeces from around the world. These data were all short read (150PE to 300PE) Illumina sequences.
2. **Published PacBio HiFi sequence:** THere is a single pacbio hifi sequence published. I will generally reference this sequence by the name **near_complete** THis sequences has good mitochondrial data!
3. **Imani Sequence:** Imani sequenced about 27 (double check number) bd isolates from across the California Sierra Nevada. THese isolates were sequenced on an Illumina Seuqnencer PE300. I think they were NovaSeq but not 100% sure, I do know that they were sequenced on one of the newer Illumina machines! As far as I am aware these data have not been published in any way.
4. **New Sequence:** I hope to sequence some number of new sequences! Number TBD but if I am lucky these will be done on a PacBio hifi sequencer!

4 NGS Analysis

4.1 General NGS Analysis Notes

Creatind a de-novo assembly from NGS sequence data is generally not super successful. It is most certainly possible however the results are generally less than desireable. When used for whole genome assembly NGS data can miss large structural variations in the genomes and might miss novel genes. It also struggles with highly repetitive regions. NGS data will, however, accurately capture SNPS at a reasonably high fidelity. The benefit of NGS data is predominently the price.

The Good news, however, is we already have a published PacBio reference genome! If we already have a good reference genome then aligning illumina sequence data especially paired end seuqnces can be EXTREMELY successful and is more likely to capture SV's although novel genenes will likely still be missed.

4.2 General steps

4.2.1 START: Pre-processed Sequencing Reads

De-multiplexed reads where individual bp calls have been made. This should include both the FWD and REV sequences of paired end reads if applicable. These reads may be long (pacbio) or short (illumina, sanger). Often this pre-processing will be performed by the sequencing platform/service providor. If this has not been performed then follow the service providords direction on cleaning and pre-processing your reads, this process is VERY specific to the sequencing platform.

4.2.2 QC:

Run all sequencing reads through some sequence quality metric tool. FastQC is common and easy to use. There are others which I shall not investigate significantly here. I used FastQC.

4.3 Quality Filter (if needed)

Hopefully there are not any reads that are significantly problematic. If there are specific reads that are bad this is a good time to toss them out. There are specific tools for this however many of the trimming tools discussed the next section can do this. Often the sequencing service will do this during de-multiplexing. All my samples had really high Phred scores so no sequences need to be removed purly based on phred score. THey do need to be trimmed though....

Note: If there is significant GC contetnt bias then you may need to drop or at least filter/compensate your sequences. Incorrect GC content (e.g. a bimodal distrobution) indicates that you have over sequenced a specific, often repetitive region of DNA. It may be wise to filter out all the reads with abnormally high

GC content and see if they are all very similar in sequence and perhaps consider blasting them to see if your sample was contaminated or if they align with a specific region. Depending on the cause of the bimodality you can filter or remove some of these samples to create a theoretically more representitive sample set. Ofcourse this is a nuanced art since there are many genomes and some may truely have a biomodal GC content distrobution, this is very unlikely but you must employ your knowledge about YOUR organism to make an accurate determination. Ultimately you will likely be wrong no matter what you do.... such is life.

4.3.1 Trim Low-quality ends:

Often the confidence in base pair calls is quite low towards the of each read. Dropping the low quality BP is generally a good idea to make sure you don't have incorrect SNP calls. THis can be done with *Trimmomatic, Cutadapt, BBduk, FASTP, Seqtk, NGSQC, SAMStat, FastQC htSeqTools, seqyclean* others?

It may not always be clear if your sequences have been trimmed. Sometimes the service provider will trim your sequences (illumina tool does this quite well). Typically if ALL of yoru reads are exactly the same length (e.g. 151 bp for 150PE reads) then the sequences likely have NOT been trimmed. Trimming typically will not remove equal number of bp from every read and thus should create a left skewed normal distrubution with an average that is 10ish bp shorter than the target read length (e.g. 140bp for 150PE). This 10ish bp decrease is mostly due to 5' bias trimming and not the adapter content bias trimming.

The trimming tool should perform 2 key tasts:

- Trip 3' Adapters. The sequencing tool will often "run-through" the end of the target dna and will sequence the adapter itself. This will cause all sorts of issues down the line. These 3' adapters should be trimmed, this is usually done by explicitly looking for the adapter sequence and then trimming EVERYTHING on the 3' side of that sequence. All sequence data after the adapter must be assumed as invalid and is tossed out. Hopefully only a moderate subset of the sequences need to have the 3' trimmed as this trimming can notably decrease your sequence k-mer length which isn't ideal. Less than may 20% needing to be trimmed is pretty great. You typically don't need to trim any ADAPTERS from the 5' end since the polymerase is running from the 5' end and starts slightly after the adapters. If you do have significant (any more than 1-5%) of reads with 5' adapter sequences then you should seriously question the sequencing technology and investigate. I have never seen this myself and frankly don't know what I would do.

- 5' Trimming. Just because you don't (or at least shouldn't) have 5' adapter sequences doesn't mean that you donot need to trim hte 5' end. If you look at all the samples that are sequenced in a single "run" then the 5' GC content you WILL see a matching pattern of about 15bp for every sequence regardless of the true sequence identity. This is caused by many reasons but often because of the fragmentation step. In theory the enzyme that chops up your dna does this at random.... however we know there are always biases so and may cut in a less than random fashon. There are many other reasons inherent to the sequening methods that also drives some of this 5' bias. Regardless it is likely best to remove the first 10-15bp of all to remove this bias. Many trimming tools will handle this just fine.

There are multiple tools that can do your adapter trimming. Unlike alignment tools they honestly all perform relatively similarly, and which specific tool is often moot especially if you have access to a significant amount of computation power. Here is a short list of trimmming tools and their benefits/drawbacks:

- Seqyclean [21]: An idea tool with lots of features and open source. This tool includes:
  - Removal of PCR-duplicates. During library amplification the same strand will be amplified multiple times which does not add any information. SeqyClean identifies two reads as "pcr-duplicates" if the first 35bps match. It's slightly more nuanced than that but thats the mostly how it do. PCR duplicates don't add information even to any CCNV analysis since in theory no two fragments should share an idnentical first 35pb. That would mean that two different copies of the same chromosomes fragmented in the exact same way... Unlikely to happen more than once or twice and once or twice isn't enough to change the CCNV analysis!! Removing the PCR-duplicates helps reduce dataset size AND will reduce bias in later analysis steps.
  - In RNA seq data it will remove poly-A/T regions. But won't do anything in DNA sequences
  - Contaminant Detection and Trimming: If you provide a list of possible contaminents then Seqyclean will perform k-mer matching and remove any contamination reads (e.g. reads that match the sequence of the provide contaminents). Contamination detection is not required for my work thus this will be skippped
  - Atapter Id and Trimming (PE overlapping): Uses SSAHA [15] algorythm adapter identification then will trim. Importantly in this step it combines paired end reads as well. In theory a paired end read should match perfectly in the middle with adapters sitting off on each reads 3' end. THis can create a high confident consensus read and sometimes will increase the length of the individual read. This step is essentially the same in AdapterRemoval and Trimmomatic!
  - Quality Trimming: Removes bp from the ends of a read taht are significantly lower than the rest of the read and that are lower than 20phred score. Has a rolling 50 and 10 bp window with a similar algorythum as Lucy.
- Trimmomatic: [2] An industry standard tool that is fast and effecient.

- Adapter Trimming: Requires a user provided list of adapter sequences to "remove" Can include any number of seqeces here (pcr producs, adapters, contamininates). Performs this trimming in one of two modes:
  1. Simple Mode: Super fast, each read is scanned from the 5' to 3' end any sequence matching a the user specified sequence. If there is an alignment everythign on the 3' end is trimmed. The user can specify the number of accepted mismatches. This will find the targeted region anywhere in the read quickly HOWEVER it will NOT detect partial adapter sequences as is common in "read through" type errors.
  2. Palindrome mode: Will find the partial adapter sequence matches at the end of a sequence (common in read through adapter errors). It does this predominently through aligning paired ends and removing the 3' overhand with some aditional user provided partial sequence mapping checks. See their publication for details [2]. This is highly beneficial and can create much longer reads than otherwise possible! Works best IF you have paired end reads (as all modern sequencing should).
- Quality Filter: Two main quality filtering methods.
  1. Sliding Window Filtering: Standard approach. Scans from 3' to 5' and when the average quality of a sliding window of bases drops below a minimum threshold those bases are dropped. This way a single weak base won't cause an early truncation of the sequence while still removing the poor sequencing results towards the end of a sequence. This runs really quickly which is rad!
  2. Maximum Information Filtering: This method tries to balance keeping long reads while dropping bad bases. Really short reads are worthless however long reads with really poor phred scores are also mostly worthless. In theory there is some tipping point where a longer sequence with imperfect base calls is better than a shorter sequence with only high confidence calls. This mode tries to find this balance to "Maximize Information" within the read.

- Cutadapt: [10] A commonly used tool that is VERY effective with Illumina data. Requires detailed understanding of command line tools. Allows simple direct adapter searches and removals. Doesn't do much automatically but allows for specific tuning of each trim which can be beneficial. This has been updated and improved through at least AUG 2025! Cutadapt will perform:
  1. Adapter Alignment and Removal. I am confused by the way it aligns and identifies the adapters. However it does successfully do this on the 3' end!

2. Quality trimming. Drop low quality bp scores from eithe just the 3' end or from BOTH the 3' and 5' ends. Tries to balance keeping some bad scores that are surrounded by good scores.
3. Will remove ployA in RNA sequences

- BBduk: [1] Debeloped by the Joint Genome Institute at LBNL in combination with DOE BBduk (Decontamination using Kmers) is a specific tool within the BBTools grouping. BBTools includes other mapping, masking , and merging tools that all work well together. This tool appears to be a state of the art cleaning tool with an impressive array of possibilities. It is hard to find information on the algorythm though.... Must consider. The more I read about BBDuk (and the rest of BBTools) the more suspicious I become. They refuse to publish a paper or produce a solid comparison on anything but simulated data. Lots of forum posts about how great it is, and it is most certainly fast and easy to use.... but I care most about accuracy and I wan't it to be good but I haven't found a paper showing how good it is yet.
  1. Quality-Trim Both ends
  2. Adapter Trim - Unsure if adapters must be specified or if a library of adapters are provided
  3. Force Trim Left and Force Trim Right: Can force trim the ends of your sequence for any reason. A common reason to trim the 5' end would be the GCAT is often not even at the beginning. Be careful though! Just cause the balance looks wrong doesn't mean that it should be trimmed. Mapping with a histogram should show if the first base pairs are frequently wrong or not. If they are not frequently wrong then don't remove those base pairs even if the GCAT frequency is off! Peep this thread post 12 https://www.seqanswers.com/forum/bioinformatics/bioinformatics-aa/37399-introducing-bbduk-adapter-quality-trimming-and-filtering?t=42776
- FASTP
- Seqtk
- SAMStat - might be a qc tool not trimming
- htSeqTools
- AlienTrimmer
- AdapterRemoval
- Skewer
- Sickle
- Ea-Utils

- Picard
- PEAR
- IeeeHom
- Lucy: Cant run paired FastQ
- Btrim: Cant run paired FastQ
- nesoni clip: A handle tool for trimming a specific number of bases of your sequence. This could be used to trim the first 10-15 bp of my sequence
- SO MANY MORE. At the end of the day there are so many tools and many perform very similarly. Best to just QC the results and make sure you have trimmed all the things that need to be and then just move on with life haha!

4.3.2 Genome Size Estimation, Read Quality Estimate:

NOTE: The method here is not the same as long read methods!! Short reads have slightly different statistics!

Helpful metrics should include: Histogram of read lengths, read depth, and others. Statistics for individual and cumulative samples should be listed. THis can help us understand if a specific sequenced isolate had problematic sequencing which could explain any erronous or wierd results!

4.3.3 Sequence Alignment:

Align sequence to reference genome. Sequence alignment is arguable the most impotant step of any whole genome analysis especially if phylogenies are to be created. Mistakes in alignment can cause quite significant changes in end results. The callange with sequence alignment is the inherent trade-off between mapping speed and the qulaity of that alignment. There is a fair argument that if you have access to a good computing cluster then sacrificing compute time for quality is recommended, do note, however, that just because a tool is slow it will not neccessarily have a higher quality. Additional consideration must be given to the characteristics of spp/genome you are trying to align. Low-complexity squences, repetitive regions, collapsed copies of sequences, contaminations, or gaps in the genome all present challenges to alignment and each tool deals with these issues in slightly different ways. Lastly the **REFERENCE GENOME** can have ENOURMOUS impacts on the end alignment quality. A low quality referenve genome with gaps, or imporperly characterized repetitive regions will wreck havoc to any alignment tool with some tools fairing better to some types of issues than others [18]. Thankfull I have the high quality **near_complete** PacBio sequnce.

Before we discuss the common tools uesed we must first mention the four primary indexing methods these tools use for multiple align.

1. **Burrow-Wheelers Transformation:** Developed in 1983 by David Wheeler and Michael Burrows the Burrow-Wheelers Transformation will reorganize a string of text grouping of similar characters together enabling faster charater comparison. Originally used for preprocessing of strings before compression it has become very helpful for alignment algorithms. If you want to explore how to perform the BWT look at Advanced Data Structures: Burrows-Wheeler Transform. In practice the BWT is performed on the reference genome, simultaniously an indexing step is porformed to create a map of each chacater in the BWT to the original position in the genome. This step can take some time but is only performed once. Once this BWT and index has been created we can take each individual short read NGS sequence and querey the BWR for the first closest match (this is covered quite extensively by DB querie scientists). The real magic is in how the algorithm handles inexact matching and gaps and is where most of the different BWT based algorithms differ (I think). I frankly cannot comment on how this process is done. Ultimately, however, the BWT based methods use very little memory after the indexing step has been completed and are extremely computationally efficient albiet with some potential loss in accuracy due to how inexact matching must be handled.

2. **Reference Genome Hash table Indexing:** NOTE: I am not expert on this and I have not researched the details of this process significantly. THis is my current understanding but you should use other resources if you want to understand the details of these processes.

   Similar to BWT hash table indexing is a database indexing method designed for faster database quiries. I am unsure when this was first developed but has tables have been around for a VERY long time. Funcitionally instead of representing a genome as a whole string the genome can be "indexed" into a hash table where each value in the hash table is a list of length k (also know as a k-mer). These k-mers are of some preselected length k (the specific length is actually super important). Typically there is some "offset" array of k-mers that repersent different starting positions of a potential k-mer which points to a specific position within some position array where the full sequence is listed. THe k-mers within the offset array can overlap significantly and contain the same information multiple times in different ways which helps with the oligemer comparison later. Once this has map has been created the piece of "target" DNA that is to be aligned is then broken down into overlapping "target" k-mers and queried against the previously created hash index. WHen matches are found the "seed location" aka the locations where the short target k-mers are recorded. Up until this point, as I understand, many hash index based systems operate very similarly. There is some deviation with long read sequences since theoretically the long read is known to go together but the basic fundamentals are approximately the same.

   Once the seeds have been determined a pleathora of possible algorithms are used to examin all the possible seeds for all providded "target" k-mers and find the overall "best" alignment. Best is arguably subjective and much care should be used when deciding which algorithm you use for a spcific species, some are better at expecting large structural variants others are better at handling small SNPS, some are better for short read (and thus short k-mer) sequences and others are better for long read (long k-mer) alignment.

A note on k-mer length. k-mer length is CRITICAL. Theoretically we need each k-mer to be whoally unique within the genome. If k is too small then there will undoutably be repeated k-mers within the hash. The Highly repetitive regions in many genomes cause a significant issue here and can easily require $k > 1000$ to have confidence in sequence alignments unless repetitive regions are dropped. Large k-mers however will be less sensitive which is not ideal. There are also two hard limitations to choice in k-mer length. 1.k cannont be longer than the sequence fragment size. E.g. if you sequences on illumina PE150 then $k <= 150$ we cannot create data from nothing. This specific limitiation is a driving factor for longer read sequencing methods like PacBio and Nanopore. 2. Large k values present a SIGNIFICANT compute challenge, the RAM required to store the hash index increases linearly with k and genome size. For a human genome of 3 billion base pairs and a $k = 100bp$ the hash table would easily need 75GB of RAM ($RAM = GenomeSize * k/4$) and thats assuming minimal overhead RAM costs for the hash. Of course there are ways to obtain this amount of RAM and the hash algorithum can be tuned to reduce RAM overhead but it is still a notable problem. Choice of k-mer length is not trivial and can change results and compute time dramatically. Choose k-mer carefully.

3. **Hashing Reads:** Theoretically similar to hashing the genome. Unsure of the nuances here

4. **Sorting and Mergin:** Unsure the implications of this method. Appears to be significantly less common than the first two. Includes Slider and Syzygy apparently [12]

TLDR: Fundamental methods of alignment: BWT is a fast effecient relatively accurate method and is especially suited to short read sequencing. Hash table indexing is memory intensive and slow but usually provides the highest accuracy and can detect large genomic rearragements and other similar genomic phenomena more easily than many BWT methods. HOWEVER, because each of these methods depend on many pre and post processing steps outcomes of any one tool VERY dependent on the specific tool. Do not select a tool simply because it is a BWT or a Hash table index, select a tool because it performs with your system and sequencing platform. [8][19][11]

Lastly: Just because a tool works REALLY REALLY well for the human genome doesn't mean it will work for your sample type. From a genomic phenomena POV the human genome is pretty darn simple

and many tools designed for human genome sequencing will fail to capture the true nuance of your system

Note: There are multiple metrics for how well some sequence aligned to the reference genome.

1. **Mapping Percentage:** The precentage of reads that were mapped to the genome. Higher is generally better. Above 90% is probably good. Importantly just because a read was mapped to the genome doesn't mean it was mapped to the right spot. This presents a significant challange.

2. **Other Metrics:** THere are countless other metrics one may be interested in, however, genetic sequencing presents an unowable problem. Rarely do we know what the "right" answer is so although we can asses each tools sensitivity and accuracy with respect to some ground tuth we have no truth to ground. Many groups have created simulated genomes to test these tools and have provided quite intersting and helpful comparisons for how these tools are likley performing. Importantly it has been noted that the performance of each tool changes based on the specific organism (or at least kingdom) that is being sequenced. In my opinion there is anough nuanced variation in these studies that if you need the utmost confidence in your results then you must either find such a study that has been performed on your system of interst or perform such a study yourself! THis is likely important if you know your organism presents some rather unique methods of inheretance or mutation. If this is not the case then just choose any of them and they will probably give you a decent answer. Better yet use a few tools and if they all agree then that answer is probably as close to right as you will get. Do also think about your final application and what specifically you are trying to learn, that can guide you if you want a more accurate or more sensitive tool.

3. **I am partial:** Currently I am moderatly inclined to use Novoalign, BWA-MEM, and STAMPY paired with the GATK variant caller. They seem to have a good balance of sensitivity and accuracy in most tests I have read.

4. Unfortunately, It appears in most cases sensitivity and accuracy are inversely related.

Here are a few of the existing alignment tools and some commentary on their best uses:

1. **Burrow-Wheeler Transformation**

    - **CLC-mapper:** CLC-mapper is developed by Qiagen. As such it is a proprietary softward suit that is primarily used through a graphical user interface. This significantly limits it's use on compute clusters and as such, in my opinion, is not suitable for large scale genomic analysis (admittadly I have never used this tool). Regardless according to [18] it does not provide the same quality results as novoalign or BWA-MEM when used to align plant genomes (human genomes are more simple and perhaps CLC-mapper is better for these more rudementary genomes) (yes I know I am throwing shade lmao - go work with a non-"model" genome and then come talk to me. Human genetics are easy).

- **SOAP2:** Uses Burrows-Wheeler Transformation indexing algorithm. [18] [20]
- **Bowtie2:** Based in the burrow-wheelers transformation Bowtie2 is an excellent longer read illumina style sequences around 150-500bp (Bowtie2 is not particularly suited for modern long read sequencing e.g. pacbio or nanopore). Bowtie2 uses a remarkably little memory and is not particularly core intensive making it an ideal platform if you do not have access to a compute cluster. It is generally more sensitive than many BWT counterparts such as BWA (Burrow-wheelers alignment) and SOAP2. I have found that more than 36 cores will not particularly decrease run times. Be sure to set proper setting for Paired End sequences and non-paired end sequences! You can find run code and some associated README in this git project! [7] [18]
- **BWA-MEM:** An update on the older BWA tool. This tool regularly comes up as one of the best BWT based alignment softwares. Combined wtih GATK BWA-MEM should be fast and give really good results for short read sequencing methods.
- **BWA-SW:** Finds local alifnmnts of the reads rather than global alignments.
- **GEM3:** [18]

2. **Reference Genome Hash Table Indexing:**

- **novoAlign:** Optimized for NGS data. Unsure if it works for long read sequencing I doubt it. Harolded as one of the most accurate alignment tools NovoAlign is based on hash table indexing. Full details of how this system works is a bit hard to find. It appears to be proprietary to some notable degree. SHould be free to institutions [18]
- **GSNAP:** Uses hash table indexing. [20]
- **SNAP:** Uses large continuous seeds
- **BFAST:** Uses large spaced seeds to improve sensitivity
- **GASSST:** Uses a large array of filters to improve alignment speeds
- **SHRiMP2:** Uses large spaced seeds to improve sensitivity
- **SeqAlto:** Uses large continuous seeds. Is only suitable to sequencing methods where reads are greater than 100bp. Published here Fast and Accurate read alignment for resequencing [12]

3. **Read Hash Table Indexing:** I don't quite understand how this differes/is better/worse from the Genome Hash Table indexing. Not super common though. Supposedly helpful when the reference genome is small (e.g. a transcriptome) [12]

   - **MAQ:**
   - **SeqMap:**

4. **Sorting and Merging:**

   - **Slider:** Requires a large fast RAM
   - **Syzygy:** Requires a large fast RAM

5. **Other/Hybrid:**

   - **Stampy:** A fast yet sensitive alignment algorythm that uses a hybrid mapping algotythum (hybrid between a hash and the burrows-wheeler method) to be more sensitive than the burrow-wheeler transformation counterparts (bowtie, bowtie2 ELAND, BWA) but faster than the high sensitivity hash based systems (Novalign, Mosaic). Stampy is particularly well suited to short read Illumina sequencing. Link to Stampy Git https://github.com/uwb-linux/stampy [9]

6. **Unknwon:**

   - **Geneious Prime:** Proprietary tool ($200 yearly). The Geneious alignment tool is easy to use and in my experience fairly accurate. FIND PAPER COMPARING ACCURACE TO OTHER TOOLS. The although GUI interface is convinient it severly limits the use of compute clusters (unless there is some geneious command line tool that I am un-aware of...) FIND THE ALGORITHM TYPE IT USES. In my experience it is substantially slower than Bowtie2 but appears to provide more accurate results and handls SVs slightly better. GEneious prime software is quite nice when trying to work with genetics datasets and undestand what you are looking at so I do recommend the software in general. However if you are trying to align more than just a few sequences then I would likely avoid this tool.
   - **BBmap:** Need to investigate but advertized as ideal for highly mutated and changing genomes.

7. **Old or depricated tools:**

   - **Bowtie:** The older brother of Bowtie2, bowtie is really only good for very short reads (¡50bp). Though a historically fantastic algorithum newer algorithums have both a higher sensitivity and increased speed.

- **BWA:** The old version of BWA. Careful of the exact name of BWA and BWA-MEM, many software tools were benchmarked against "BWA" and it is not always specific if they benchmarked agains the most modern version "BWA-MEM". Frankly, neither BWA nor BWA-MEM is the best tool for my application so the issue is largely moot!

Some advertized "alignment" tools should not be used in this short read sequence alignment steps:

1. **Mauve:** A powerful too for multiple alignments. Mauve is designed to detect whole genome rearrangments and is bessed used for comparative genomics. Aka taking two assembled (likely de-novo assembled) genomes and compairing them! Helpful for answering questions about synteny, or identification of large SVs. This can be an excellent tool for multiple alignment prior to building a phylogeny. It cannot handle short read sequences. Also note that Mauve has a function (if rudimentary) GUI which makes it slightly less convinient to run on a cluster. I belive Mauve alignment is possible using only command lind tools and associated run scripts however I have not yet attempted this (Jan 2026).

2. **LastZ:** A good tool for long sequences LastZ is more similar to Mauve than the tools we ned fo this step. LastZ is an excellent choice for multiple alignment prior to building a phylogony!

Discussion on the tools I plan to try: Currently it is not possible to know which tool will give me the best results thus I hope to try: **BWA-MEMs, STAMPY, bowtie2, Novoalign, and maybe SeqAlto**. With the known CCNV in Bd I have concerns of repetitive areas and large structural variants so I want something that can handle these variants well, I believe that Novoalign, STAMPY, and SeqAlto are particularly suited to these challanges. Most of the NGS data I have from Imani are PE150 or larger so having a longer k-mers on the order of 28 or so will give me decent coverage without sacrificing specificity. For some of the downloaded historical reads I should decrease the k-mer length notably. In my case I don't really need to worry about memory since I have access to the cluster and the Bd genom is pretty small. I am particularly inclined to use STAMPY because it has been used with good results in Bd in the past and according to Mu et al. 2012 [12] Stampy delt with indels, SNPs, and MNPs very remarkably. Novoalign has also been updated since then and has some really good reviews dispite the claim that it can run quite slow. Lastly I want to use BWA-MEMs because it has been used a good bit in Bd genetics research and has proven itself accurate. Bowtie2 is on the list because it is easy to impliment and fast to run. It would be nice to know if bowtie2 was adequate since using that tool would save other the required use of a cluster.

NOTE: Although I tried to make sure I cited approriate journals/publications in this list. The following publications were all extremely helpful for identifying and understanding the different tools. [6], [17], [13], [14], [4], [16], [9], [7], [6], [20], [8],[19][11], [12]

5 PacBio Analysis

5.1 Generaly pacbio analysis notes

: Creating a de-novo assembly from long read PcaBio sequencing is a really great idea! We have the data and while performing the de-novo assembly we are more likely to identify SV's and new genes. Since we already have a hifi pacbio sequence I recommend doing a hybrid type assembly where we do a de-novo assembly and then scaffold this genome to a good reference genome. At which point we can make our structural variant calls and gene annotations (see a Begginers Guid [6]).

6 What I have done (in rough order with dates)

6.1 Genious Stuff

I did some random things in Geneious including making some aligments to the genome and some rough phylo trees. I hadn't cleaned anything yet though so that was bad!

6.2 Bowtie alignment 1

Aligned stuff with bowtie using the cluster. HJadn't cleaned stuff yet though so that was bad

6.3 Illumina Sequencing Analysis Real:

- (2026-01-21: Fast QC) Analized the sequences (imanis and downloaded) with FastQC to asses the common issues. Most are pretty okay.
  - Imani's sequences
    1. Per Base sequence content: Most Failed. The first 15ish BP of most seqhences have a poor pattern of relative GC content. This pattern almost perfectly aligns with each sequencing run (5720 and 5912). All the 5720 look the same and all the 5912 look the EXACTLY the same except for 5912-JEL272 which looks somewhat like the other 5912 samples but not exact, this may be driven by its POOR GC graph which is notably bimodal.... Discussed later ULtimately I believe that the first 15ish BP should be trimmed b/c there is so much concensus between all the fragments that either it is a fragmentation bias or a bias of the specific sequencer. I suspect it is a sequences (perhaps adapter) based bias since the per bas sequence content is paired between each run.
       Be careful though! Just cause the balance looks wrong doesn't mean that it should be trimmed. Mapping with a histogram

should show if the first base pairs are frequently wrong or not. If they are not frequently wrong then don't remove those base pairs even if the GCAT frequency is off! Peep this thread post 12 https://www.seqanswers.com/forum/bioinformatics/bioinformatics-aa/37399-introducing-bbduk-adapter-quality-trimming-and-filtering?t=42776

2. Adapter Content: The adapter content for all samples is rather high for the nextera transposase sequence. Perhaps 20% at position bp position 140. This is likely because of there hasb't been any adapter trimming. That must be trimmed! Which is my next step. This trimming thing begins sometimes as soon as 50-60bp. This isn't ideal but not that big of a deal if this is relatively rare in the sample. This is caused by sequence "read-through" and really isnt so bad so long as the 3' end is trimmed. Unltimately it is caused byu inconsistent fragmentation which is kinda common in nextera kits. PolyG and polyA are other common artifacts in illumina sequencing and can/should be trimmed. If it's only a small fraction of the reads it shouldn't cause any issue at all.
3. Deduplication Level. All samples have a non "ideal" deduplication level. I am not concerned because of bds weird CCNV genome I expect there to not be even coverage across the genome.
4. 5912-JEL272: Bimodal per sequence GC content. Should be normally distrubuted around like 40% or so.
5. 5912-JEL216: Bimodal per sequence GC content. Should be normally distrubuted around like 40% or so.
6. I may need to toss JEL272 and JEL216. The bimodal GC distrobution has me sorta concerned. But so does the distro of the published genomes so IDK. Might isolate the high GC% sequences and blast those to see if the align with anythign.... Hard to say.
7. Average Length: All samples are VERY consistently 151bp in length. This is great means the machine didn't have any systemic issue. This consistent read length will change once I trim but that is to be expected. Lets trim.

   – Published sequences
     1. Hello world

- (2026-01-27 Trimmomatic:) I ran Trimmomatic version 0.40 (most recent of Jan 2026) on all of Imanis samples. I used my bd genetics analyses scripts specifically the run files within the Trimm0matic folder of this git branch.

```
# Github link: https://github.com/malloc3/bd_genetics_analysis.git
# commit 2bd99208b569fa5eb02994aa11b892a83cf0aa78 (HEAD -> main, origin
#   Author: Cannon Mallory <mallorycannon15@gmail.com>
#   Date:   Tue Jan 27 19:33:04 2026 -0800
```

```
#
#    editing the error and outlog names
```

The Specific settings used for the trimmomatic command were

```
java -jar "./Trimmomatic/target/trimmomatic-0.40.jar" PE -threads $numb
-trimlog "$trim_log_file" -summary "$summary_file" -compressLevel 9 -ve
"$full_input_file_path_1" "$full_input_file_path_2"\
"$full_output_file_path_1_paired" "$full_output_file_path_1_unpaired" \
"$full_output_file_path_2_paired" "$full_output_file_path_2_unpaired" \
HEADCROP:15\
ILLUMINACLIP:Trimmomatic/adapters/NexteraPE-PE.fa:2:30:10 \
LEADING:3 \
TRAILING:3 \
MAXINFO:150:5 \
MINLEN:36

# HEADCROP:15 removes the first 15 bases without question (I htink I ne
#ILLUMINACLIP:  NexteraPE-PE.fa: :<seed mismatches>:<palindrome clip th
#       <simple clip threshold>:<minAdapterLengthPalindrome>:<keepBothR
#   ## NexteraPE-PE.fa - is the folder that has the adapters we want to
#                   The specific format of this folder is IMPORTANT bey
#   ## seedMismatches: specifies the maximum mismatch count which will
#   ## palindromeClipThreshold: specifies how accurate the match betwee
#   ## simpleClipThreshold: specifies how accurate the match between an
#   ## minAdapterLengthPalindrome: (optional int) specifies the minimum
#   ## keepBothReads: (optional boolean) specifies if both reads should
#
#
#  MAXINFO uses their special algorthm to maintain maximum read length
#                Its clever and should do good for us
# Remove leading low quality or N bases (below quality 3) (LEADING:3)
# Remove trailing low quality or N bases (below quality 3) (TRAILING:3)
# Scan the read with a 4-base wide sliding window, cutting when the ave
#
# Drop reads below 36 bases long (MINLEN:36)
```

All samples retained over 95% of all reads wtih most retaining 98% to 99% of all reads post trim.

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